

E9 205 Machine Learning for Signal Processing

**Probabilistic PCA and Non-linear
Regression**

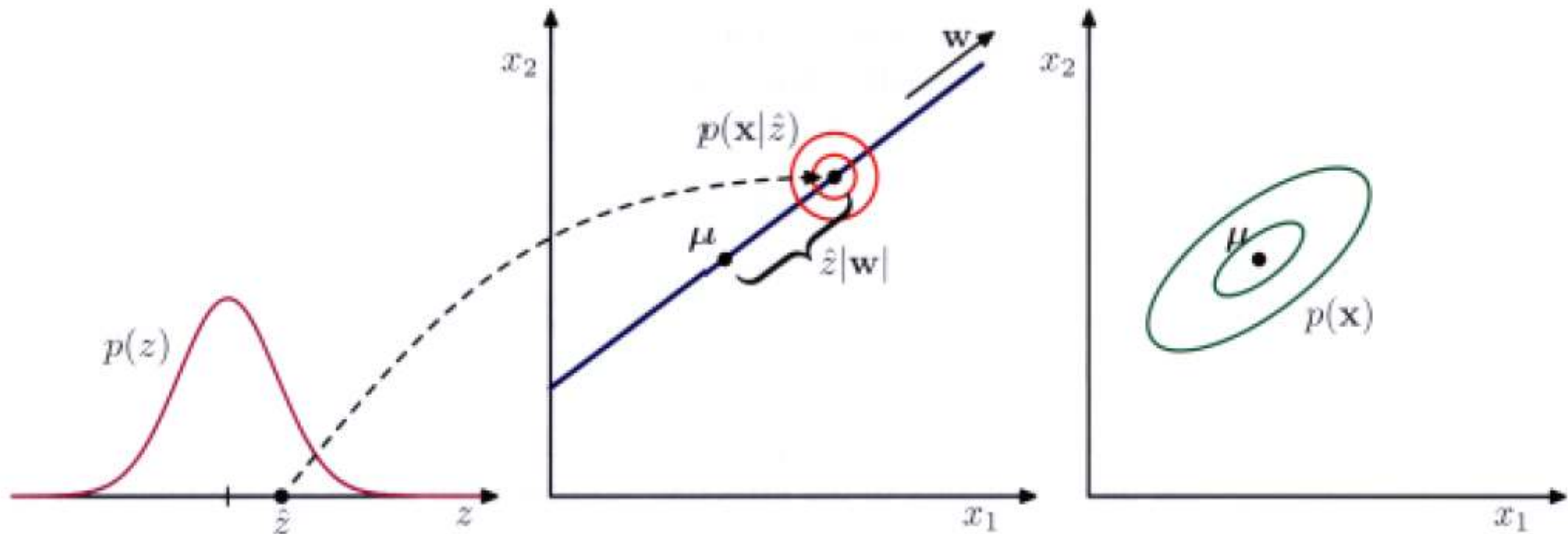
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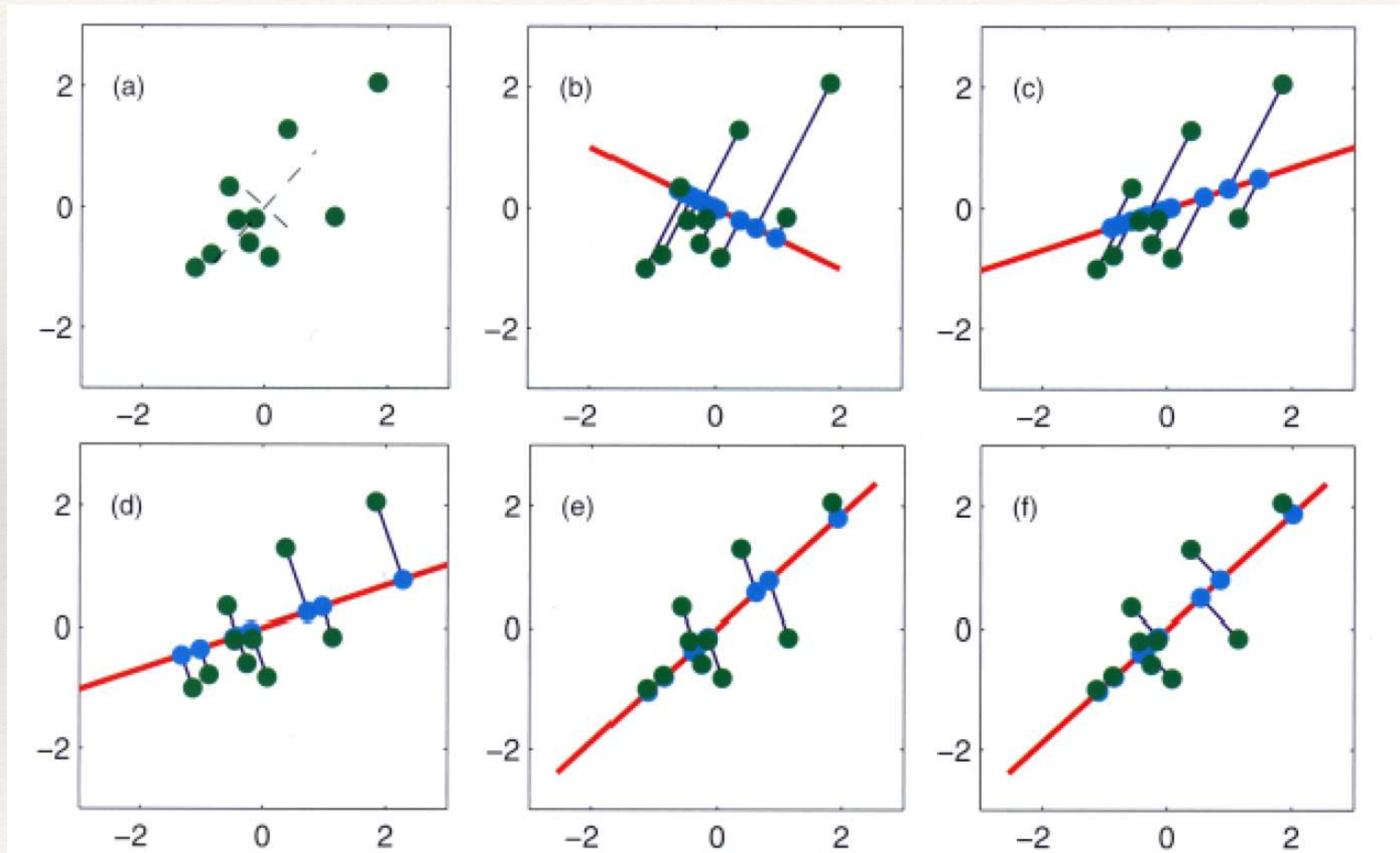
Probabilistic PCA



$$p(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}|\mathbf{W}\mathbf{z} + \mu, \sigma^2\mathbf{I})$$

$$p(\mathbf{z}) = \mathcal{N}(\mathbf{z}|\mathbf{0}, \mathbf{I}).$$

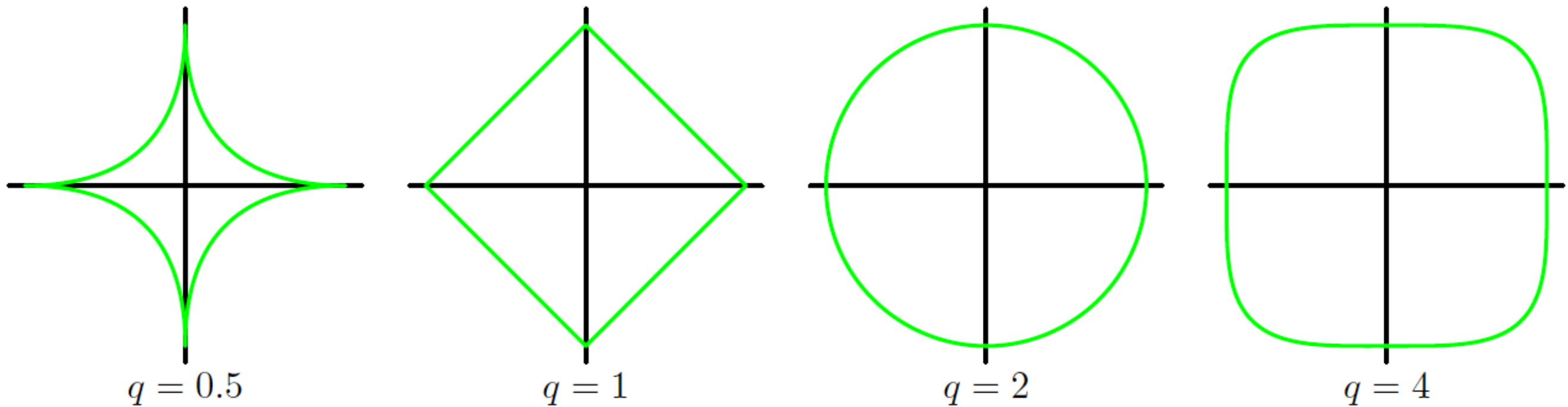
Probabilistic PCA - EM Algorithm



Probabilistic PCA

- Missing Data Problems
- Probabilistic Interpretation of PCA
- Useful for Classification

Regularization in Least Squares



$$\frac{1}{2} \sum_{n=1}^N \{t_n - \mathbf{w}^T \phi(\mathbf{x}_n)\}^2 + \frac{\lambda}{2} \sum_{j=1}^M |w_j|^q$$

Regularization in Least Squares

